

Stages of a solar eclipse

Time-lapse photography shows how the Moon covers the Sun in stages. In a total eclipse, the Sun is completely covered – this is called totality – for a few minutes. The Sun's outer atmosphere, the corona, can be seen clearly at this time.

There is one total solar eclipse every one to two years, and, very rarely, two in one year. The next time two solar eclipses occur in the same year will be in 2057.

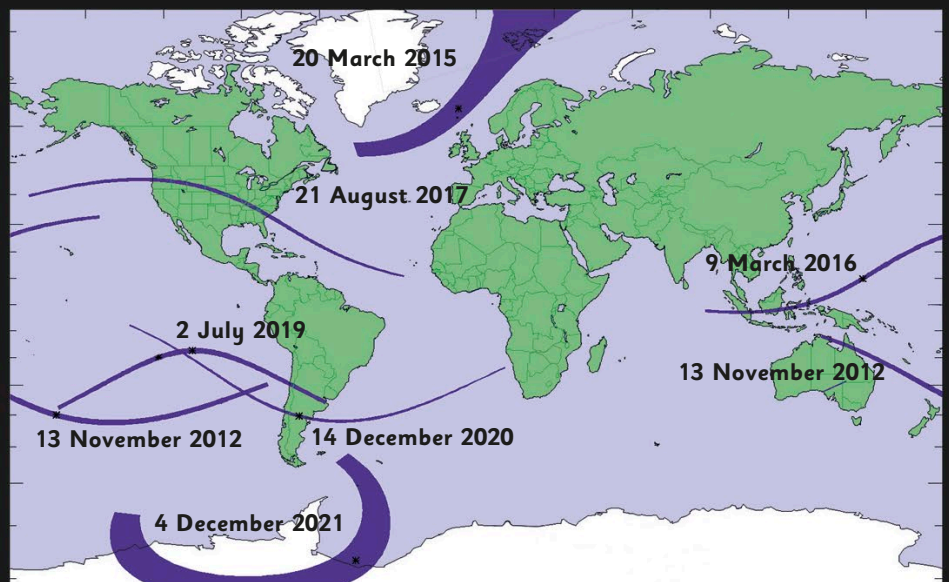
It takes about one hour for the Moon to block the Sun's light, once it begins to move across the Sun.

Ringed wonder

In the instant before the Sun disappears behind the Moon, sunlight sometimes streams between mountains on the Moon's surface, producing a stunning effect known as a diamond-ring.

The diamond-ring effect lasts for just a few seconds.

The map shows the paths of total solar eclipses from 2012 to 2021.



Plotting eclipses

Total solar eclipses occur once every 15 months or so, and maps such as these are used to plot the path of future eclipses. The shadow from a total eclipse follows a narrow path and often falls on an ocean – so it won't be seen (unless you're on a boat!).